

# emotion

## 1.Characterizing Emotion

针对特定的身体和社会挑战和机会而发生的、片段性的、相对短期的、基于生物感知、体验、生理、行为和沟通模式。

## 2.Theories of emotion:

### 1) Evolutionary Theories of Emotion 进化学理论

Evolved and Adaptive, Separate, Distinct physiology, Universal, Innate

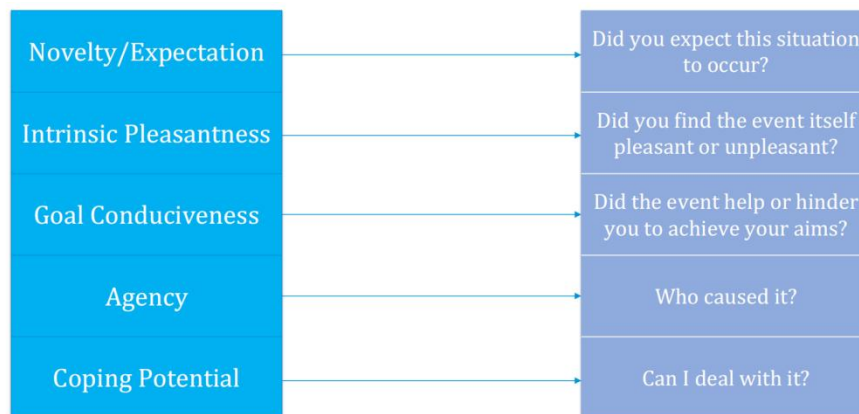
**Causes:** Emotions caused by activation of Affect Program

**Biological givens:** Basic emotions are biological givens

**Components:** Facial expressions, behavior (action tendencies), physiology, changes to cognition; Subjective experience less central component

**Integration:** High integration between components

### 2) Appraisal Theories of Emotion 认知评价理论



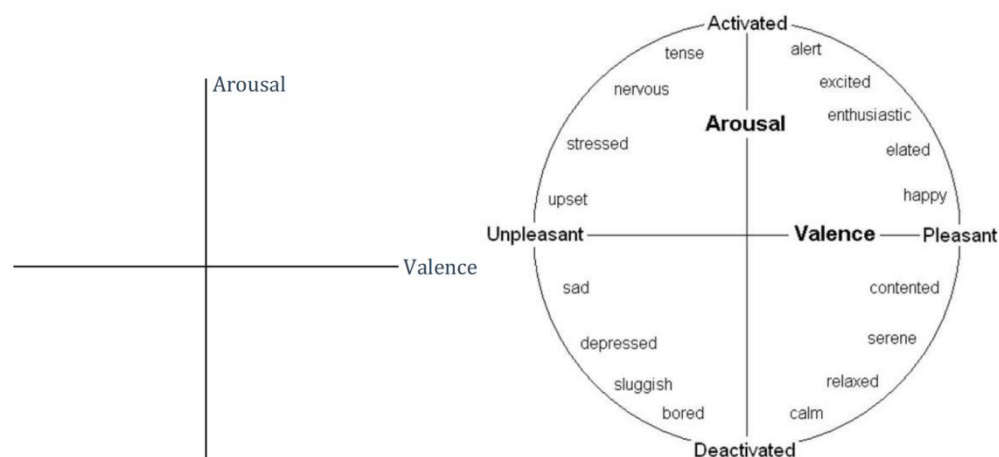
**Causes:** Emotions caused by appraisals

**Biological givens:** Novelty and valence (at least) are biological givens

**Components:** Appraisals can affect other components, but not necessarily all together

**Integration:** Less strong integration between components

### 3) Constructionist Theories of Emotion 情绪建构论



**Causes:** Emotions caused by valence and arousal (core affect) in combination with conceptual knowledge

**Biological givens:** Core affect is a biological given

**Components:** Components vary across experiences; Subjective experience central

**Integration:** Little integration between components(because of variable)

### 3.How to Study Emotions?

#### Inducing Emotions

##### 1) Pictures

Validated stimulus sets; Induces weak, short-lived emotions; Good for multiple measures; Works only for some emotions

##### 2) Movies

Validated stimulus sets; Can elicit specific emotions; Can elicit strong emotions; Few repetitions; Works only for some emotions

##### 3) Music

Validated stimulus sets; Induces diffuse emotions or moods; Works only for some emotions

##### 4) Memories

Can elicit intense emotions; Can elicit specific emotions; Few repetitions; Works for most emotions

##### 5) Scripted social interactions

Can elicit intense emotions; Can elicit specific emotions; Typically only one trial per participant; Works only for some emotions

#### Measuring Emotions

##### 1) Self-report (questionnaires)

##### 2) Biological measures

Skin Conductance (EDA/GSR)

Brain Activity (EEG, fMRI)

Heart Rate (ECG)

##### 3) Nonverbal expressions

Facial Expressions

a) Facial EMG

b) Automatic FACS-based Software: FaceReader, iMotion,

c) Manual FACS Coding

Pupil Dilation (eye tracking)

Gesture

Touch

Sympathy, Gratitude, Love, Anger, Fear, Disgust

##### 5) Voice

a) Vocalization (叹词)

b) Speech Prosody (韵律)

#### 4. Facial Expressions of Emotion

How many facial expressions of emotion are there? 28 in the United States

Facial muscles

面部动作编码系统 (Facial Action Coding System, FACS) 44 个面部动作单元

| Upper Face Action Units                                                             |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| AU 1                                                                                | AU 2                                                                                | AU 4                                                                                | AU 5                                                                                | AU 6                                                                                | AU 7                                                                                 |
|    |    |    |    |    |    |
| Inner Brow Raiser                                                                   | Outer Brow Raiser                                                                   | Brow Lowerer                                                                        | Upper Lid Raiser                                                                    | Cheek Raiser                                                                        | Lid Tightener                                                                        |
| *AU 41                                                                              | *AU 42                                                                              | *AU 43                                                                              | AU 44                                                                               | AU 45                                                                               | AU 46                                                                                |
|    |    |    |    |    |    |
| Lid Droop                                                                           | Slit                                                                                | Eyes Closed                                                                         | Squint                                                                              | Blink                                                                               | Wink                                                                                 |
| Lower Face Action Units                                                             |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |
| AU 9                                                                                | AU 10                                                                               | AU 11                                                                               | AU 12                                                                               | AU 13                                                                               | AU 14                                                                                |
|    |    |    |    |    |    |
| Nose Wrinkler                                                                       | Upper Lip Raiser                                                                    | Nasolabial Deepener                                                                 | Lip Corner Puller                                                                   | Cheek Puffer                                                                        | Dimpler                                                                              |
| AU 15                                                                               | AU 16                                                                               | AU 17                                                                               | AU 18                                                                               | AU 20                                                                               | AU 22                                                                                |
|    |    |    |    |    |    |
| Lip Corner Depressor                                                                | Lower Lip Depressor                                                                 | Chin Raiser                                                                         | Lip Pucker                                                                          | Lip Stretcher                                                                       | Lip Funneler                                                                         |
| AU 23                                                                               | AU 24                                                                               | *AU 25                                                                              | *AU 26                                                                              | *AU 27                                                                              | AU 28                                                                                |
|  |  |  |  |  |  |
| Lip Tightener                                                                       | Lip Pressor                                                                         | Lips Part                                                                           | Jaw Drop                                                                            | Mouth Stretch                                                                       | Lip Suck                                                                             |

肌肉动作单元(Action Unit)

dynamic /static facial expressions

#### Decoding Facial Expressions

Manual FACS Coding 很难

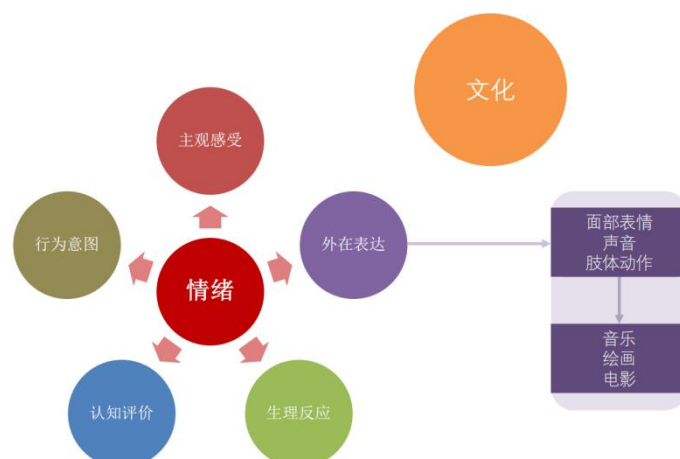
Do faces always tell the truth about feelings ?

Automatic FACS Coding

识别情绪种类和面部动作有限；比如 FaceReader 目前只能判定 7 种表情，20 个面部动作。

对自然情境下的自发表情识别信度和效度较低。

The question of universality (cross culture)



“Basic” facial expressions? (6 or 4?)

a) Anger ; b) Disgust ; c) Fear ; d) Sadness ; e) Surprise ; f) Happiness

Emotional Dialects : Individuals are better at recognizing facial expressions of emotion from their own culture than from another culture.

Compared to Westerners, Easterners are influenced more by the concurrent emotional context when judging others’ facial expressions.

## **Attitudes and Behavior**

### **1. What are attitudes?**

An attitude is an evaluation of an object along a positive-negative dimension.

#### **1) Components of attitudes**

Affect/Emotion (A)

Nearly every object triggers some degree of positive or negative emotion, which constitutes the affective component of the attitude somebody has toward it.

Cognitions (C)

Includes knowledge and beliefs about the object, as well as associated memories and images.

Behaviors (B)

The affective evaluation of good versus bad is connected to a behavioral tendency to either approach or avoid.

#### **2) How do we measure attitudes?**

##### **Example of Explicit Measures**

- A Likert Scale: A numerical scale used to assess people’s attitudes; a scale that includes a set of possible answers with labeled anchors on each extreme.

##### **Accessibility - Response Latency**

The accessibility of the attitude — how readily it comes to mind

Measure: Response Latency: The time it takes a person to respond to an attitude question

##### **Centrality**

The centrality of the attitude to the person’s belief system

Measure: A variety of attitudes within a domain and calculate how strongly each one is linked to the others

##### **Implicit Measures**

Physiological Measures

EEG (brain activity)/ fMRI (brain activity) / ECG (heart rate) / Eye tracker (eye movement)

Nonverbal Behavior Measures

Facial Expressions / Gestures

Reaction Time Measures

**Implicit Association Test (IAT)** 内隐联想测验

### **2. The attitude-behavior link**

Do attitudes determine behaviour?

## 1) Attitudes to behavior

Reasons why attitudes sometimes do not predict behavior

① There are other powerful determinants of behavior and other conflicting attitudes.

- Conflicting attitudes towards other issues
- Other powerful determinants
  - Habits / Situation
- Perceived norms of appropriate behavior

② Mismatch between attitudes and behaviours in automaticity level

Example - **Dual Attitudes (Gut Feelings vs. Analysis)** 双重态度 直觉 vs. 分析

③ Mismatch between attitudes and behaviours in specificity level.

- General attitudes predict behaviours in general
- Specific attitudes predict specific behaviours

④ Attitudes differ in strength.

- Strong attitudes are more accessible. More accessible attitudes direct behavior

## 2) Behavior to attitudes

Why and when do behaviors change attitudes?

Cognitive dissonance theory / Self-perception theory

### 3. Cognitive dissonance theory 认知失调理论

The theory that inconsistency between a person's thoughts, sentiments, and actions creates an aversive emotional state (dissonance) that leads to efforts to restore consistency.

- Festinger argued that dissonance reduction takes place only after an irrevocable decision has been made.
- Later research suggests that the same sorts of rationalization and distortion that occur after people make a decision also subconsciously take place before they make the decision.
- Once people develop a slight preference for one option over the others, they distort subsequent information to support their preference.

When----You pay a high price for something (money, time, or effort), and it turns out to be disappointing.

**Effort Justification:** The tendency of devoting mental energy to justifying what you've done.

Sweet lemons rationalization ; the IKEA effect

When----people are induced to behave in a manner that is inconsistent with their beliefs, attitudes, or values.

**Induced Compliance:** Subtly compelling people to behave in a manner that is inconsistent with their beliefs, attitudes, or values in order to elicit dissonance and therefore a change in their original attitudes and values.

Induced Compliance and Attitude Change

Induced Compliance and Undesired Behavior: "Forbidden toy" paradigm

#### 4. Self-perception theory 自我知觉理论

The theory that people come to know their own attitudes by looking at their behavior and the context in which it occurred and inferring what their attitudes must be.

- Weak and ambiguous

#### Explanation of Dissonance Effects

- **Self-Perception Theory** (Bem, 1967):
  - People aren't troubled by any unpleasant state of arousal like dissonance; they merely engage in a rational inference process.
  - They don't change their attitudes; rather, they infer what their attitudes must be.
- **Cognitive Dissonance Theory** (Festinger, 1957):
  - Inconsistency between a person's thoughts, sentiments, and actions creates an aversive emotional state (dissonance) that leads to efforts to restore consistency.

#### Reconciling

- Dissonance (Aversive Emotional State)
  - Self-report measures
  - Facial EMG
- Both might occur.
  - Dissonance reduction processes are activated when behavior is inconsistent with preexisting attitudes that are clear-cut and of some importance.
  - Self-perception processes are invoked when behavior conflicts with attitudes that are relatively vague or of less importance.

#### Persuasion

The process by which a message induces change in feelings, attitudes, or behaviors.

##### 1. Dual-Process Approach to Persuasion

##### Elaboration Likelihood Model (ELM)

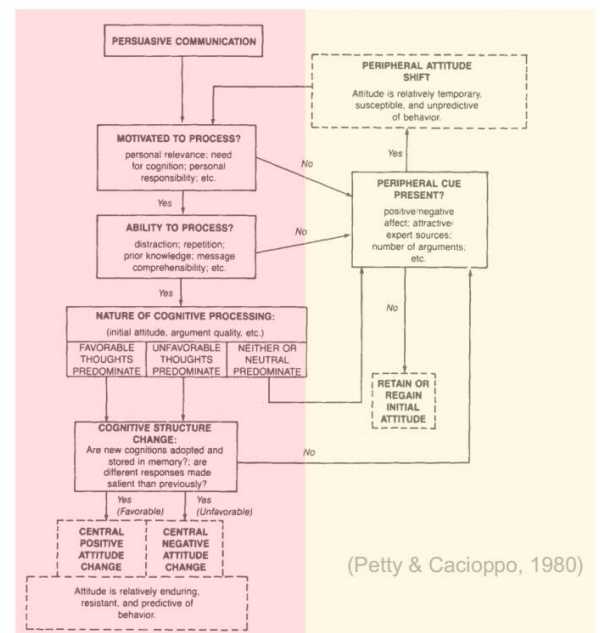
A model of persuasion maintaining that there are two different routes to persuasion: the central route(中央路径) and the peripheral route(外围路径).

Persuasion via the Central Route:

A route to persuasion wherein people think carefully and deliberately about the content of a persuasive message, attending to its logic and the strength of its arguments, as well as to related evidence and principles.

Persuasion via the Peripheral Route:

A route to persuasion wherein people attend to relatively easy-to-process, superficial cues related to persuasive message, such as its length or the expertise of attractiveness of the of source of the message.



## 2. The Elements of Persuasion

**Motivation:** Motivation to process the arguments

Personal relevance: goals, interests, or well-being

**Ability:** Ability to process the arguments

Individuals:

Cognitive resources

Intelligence/knowledge

Stimuli:

Message clarity

Limited time

### Central Route

|              | Motivated: Yes | Motivated: No |
|--------------|----------------|---------------|
| Ability: Yes | Central        | Peripheral    |
| Ability: No  | Peripheral     | Peripheral    |

**Argument quality** is important to the Central Route (related to the attitude-behavior link)

- Strong arguments – make you more favorable toward the object/issue
- Weak arguments – make you less favorable toward the object/issue

Attitudes changed via the Central Route are **Stronger Attitudes**:

- Last longer
- More predictive of behavior
- More resistant to change

### Peripheral Route

**Argument quality** is less important to the Peripheral Route : Strong arguments and weak arguments can have the same effect.

- Strong arguments are less impactful.
- Weak arguments are less damaging.

Attitudes changed via the Peripheral Route are **Weaker attitudes**:

- last less longer; more temporary
- less predictive of behavior
- less resistant to other people trying to influence our attitudes

### Yale Attitude Change Approach

#### ① Sender's Characteristics

- Credibility (Expertise, Trustworthiness/Self-Interest)
- Attractiveness (Physical Appeal, Similarity)
- Certainty/confidence

## ② Message Characteristics

- Message quality (Conclusions are explicit in the message, Sources argue against their own self-interest.)
- Vividness
- Emotion (positive feelings, fear)
- Culture – fit norms and values of culture

## ③ Audience Characteristics

- Motivation and Ability
- Age
- Mood
- Need for Cognition/need for closure: A personality variable reflecting the extent to which people engage in and enjoy effortful cognitive activities.

## 3. Metacognition and Persuasion

Metacognition: Secondary thoughts that are reflections on primary thoughts (cognitions).

- Self-Validation Hypothesis 自我证实假设

The idea that feeling confident about our thoughts validates those thoughts, making it more likely that we'll be swayed in their direction.

- Embodiment and Confidence 具身认知和自信程度

Our confidence can also be conveyed through nonverbal cues.

## 4. Resistance to Persuasion

- Attentional bias and resistance

People are inclined to attend selectively to information that confirm their original attitudes.

- Previous commitments and resistance

Many persuasive messages fail because they can't overcome the target audience's previous commitments.

- Knowledge and resistance (ELM)

People with a great deal of knowledge are more resistant to persuasion.

## Attitude Inoculation

Small attacks on people's beliefs that engage their preexisting attitudes, prior commitments, and background knowledge, enabling them to counteract a subsequent larger attack and thus resist persuasion.

## Social Influence

The many ways people affect one another, including changes in attitudes, beliefs, feelings, and behavior resulting from the comments, actions, or even the mere presence of others.

### 1. Conformity 一致性

Changing one's behavior or beliefs in response to explicit or implicit pressure (real or imagined) from others.

**Chameleon Effect** 变色龙效应



The nonconscious mimicry of the mannerisms of an interaction partner.

- The mere perception of another's behavior can automatically increase the likelihood of engaging in that behavior oneself. This is based on the fact that the brain regions responsible for perception overlap with those responsible for action.
- To facilitate smooth, gratifying interaction and to foster social connection.

**Informational Social Influence** : The reliance on other people's comments and actions as an indication of what's likely to be correct, proper, or effective

#### Sherif's Conformity Experiment

**Normative Social Influence** : The influence of other people that comes from the desire to avoid their disapproval and other social sanctions (ridicule, barbs, ostracism).

#### Asch's Conformity Experiment

Factors Affecting Conformity Pressure

- Group size • Group unanimity • Anonymity • Expertise and status • Culture
- Tight and loose cultures • Gender

### 2. Compliance 顺从

Responding favorably to an explicit request by another person.

- Reason-Based Approach

Reciprocity Norms 互惠原则

Reciprocal Concessions 以退为进

Foot-in-the-Door Technique 得寸进尺

- Emotion-Based Approach

Positive moods

o Our mood colors how we interpret events.

o Mood maintenance

Negative moods: guilt

- Norm-Based Approach

Descriptive norms

Prescriptive norms(injunctive norm)

### 3. Obedience to Authority 服从

In an unequal power relationship, submitting to the demands of the person in authority.

#### Milgram's (1963) Study

- 1) Emotional Distance of the Victim (learner) from the Subject (teacher)
- 2) Closeness of the Authority
- 3) Legitimacy of the Authority
- 4) Group influence

#### Stanford Prison Study

Social Role : Shared expectations by group members about how particular people in the group ought to behave.

#### **Deindividuation**

The loosening of normal constraints on behavior when people are in a group, leading to an increase in impulsive (and deviant) acts.

Groups can:

- make people feel less accountable for their actions (e.g., soccer hooligans)
- decrease self-awareness
- make people more likely to obey group norms

Johnson and Downing's (1979)

## Deindividuation and Social Norms

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Two Independent variables:

- **Social cues**
  - Everyone is asked to wear white outfits.
  - Prosocial cues – asked to put on nurses outfits
  - Anti-social cues – asked to put on KKK outfit
- **Deindividuation**
  - Told no way to identify who gave what shocks and no name tags
  - Told other group members could see their selection of shock levels and wore name tags

### Resisting Social Pressure

- Reactance (Brehm & Brehm, 1981) 反弹

An attempt to restore one's sense of freedom

- Flexible Correction Model (Wegener & Petty, 1997) 讲好的 意识到影响 反而会写不好的

Based on our naïve theories of influence, we correct for:

The direction of perceived influence / The extent of perceived influence

Mood and Life Satisfaction

Context and Evaluations of Kansas

## Groups

A collection of individuals who have relations to one another that make them interdependent to some significant degree.

1. The Nature and Purpose of Group
2. Functions of Groups: Social Facilitation 社会促进效应

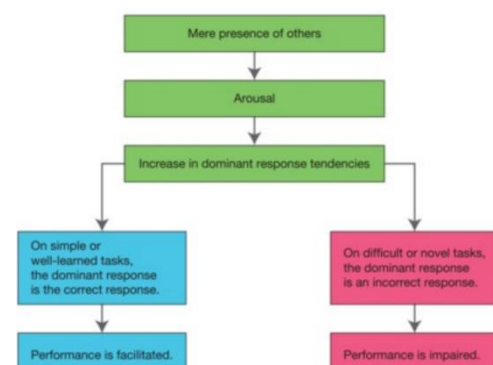
Initially a term for enhanced performance in the presence of others; now a broader term for the effect, positive or negative, of the presence of others on performance.

Zajonc's Theory of Mere Exposure

- Arousal
- Dominant response:

In a person's hierarchy of possible responses in any context, the response that person is most likely to make.

- Easy vs. Difficult



### 3. Group Decision Making 群体决策

#### **Group Polarization**

- The tendency for group decisions to be more extreme than those made by individuals; whatever way the group as a whole is leaning, group discussion tends to make it lean further in that direction.
- Strengthening of the members' average tendency.